

Premium E-Book Reader Platform

Product Requirements Document (PRD) • Version 1.0

1. Executive Summary

Build a premium React-based eBook reading platform that rivals Apple Books and Kindle while offering a handcrafted editorial aesthetic, immersive reading experience, and offline-first architecture.

2. Product Vision

Create the most delightful browser-based reading experience with typography-first design, minimal distractions, and seamless cross-device continuity.

3. Objectives

- Reading-first UX
- Beautiful Light/Dark/Sepia/AMOLED themes
- Accessibility-first
- Production-ready React architecture
- PWA with offline support

4. Target Users

Students, researchers, professionals, casual readers, book clubs, and digital libraries.

5. Tech Stack

React, Vite, TypeScript, Tailwind CSS, Framer Motion, Zustand, React Router, TanStack Query, shadcn/ui, Radix UI, IndexedDB, Service Workers, PWA.

6. Design Principles

Human-crafted UI, editorial spacing, premium typography, calm color palette, invisible UX, meaningful motion, zero clutter.

7. Information Architecture

Landing, Library, Categories, Collections, Continue Reading, Book Details, Reader, Notes, Highlights, Search, Analytics, Settings, Profile.

8. Reader Experience

Custom fonts, font sizing, margins, line height, chapter navigation, TOC, bookmarks, notes, highlights, search within book, fullscreen, focus mode.

9. Premium Features

- Optional realistic page-turn animation
- Circadian automatic theme switching
- Paper, Cream, Sepia, Charcoal and AMOLED themes
- Interactive 3D Bookshelf

- Color-coded highlights with tags
- Reading streaks and analytics
- Reading completion forecasts
- Cross-device reading synchronization
- Keyboard shortcuts inspired by VS Code & Arc
- Installable PWA
- Offline library
- Background synchronization
- Service worker caching

10. Accessibility

WCAG 2.2 AA, keyboard navigation, screen reader support, reduced motion, semantic HTML, focus indicators.

11. Performance

Lazy loading, virtualization, image optimization, code splitting, optimistic updates, cache-first strategy, <2s first load target.

12. Component Library

Book Card, Shelf, Reader Toolbar, Theme Switcher, Progress Ring, Highlight Panel, Notes Panel, Command Palette, Analytics Cards.

13. Folder Structure

Feature-based architecture with reusable hooks, services, stores, UI components, layouts, utilities and testing.

14. Roadmap

Phase 1 Foundation • Phase 2 Reader • Phase 3 Library • Phase 4 Sync • Phase 5 Analytics • Phase 6 AI Enhancements.

15. Acceptance Criteria

All interactions are responsive, animations remain smooth at 60 FPS, accessibility score >95, Lighthouse Performance >95, production-ready codebase.

Appendix

This PRD serves as the baseline specification for implementing the complete Premium E-Book Reader Platform in React.

Area	Goal
Performance	95+ Lighthouse
Accessibility	WCAG AA
PWA	Offline Reading
Animation	60 FPS
State	Zustand
UI	Editorial, handcrafted

Platforms	Desktop, Tablet, Mobile
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